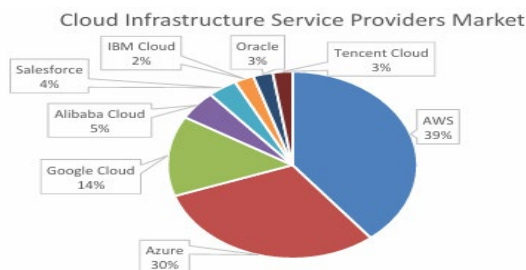


The following pages contain an artifact sample for **Goal 1b Written Communication** and **Goal 2 Critical Reasoning**.

MICROSOFT CORP (MSFT)

Overview

Microsoft Corporation is a global leader in enterprise software, cloud infrastructure, and artificial intelligence. Headquartered in Redmond, WA, and a key component of the S&P 500 and Dow Jones indices, Microsoft operates through three primary segments: Productivity and Business Processes, Intelligent Cloud, and More Personal Computing.



Its partnership with OpenAI and the integration of Copilot across Office and Dynamics have positioned Microsoft at the forefront of generative AI. Azure remains a core growth engine, supported by strong enterprise demand and increasing AI workloads. Strategic investments in GPUs, data centre capacity, developer tools, and disciplined capital allocation continue to reinforce Microsoft's competitive edge and long-term growth trajectory.

Strategic Outlook

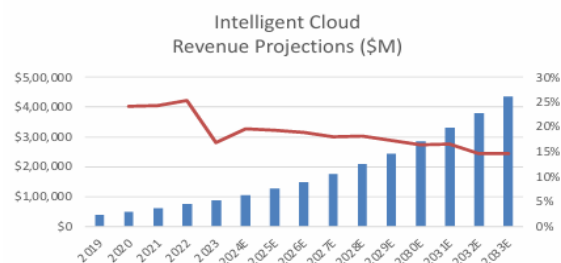
Microsoft stands at the intersection of transformative AI innovation and enterprise-scale cloud computing. Its early-mover advantage through the OpenAI partnership, combined with broad Copilot integration across productivity tools, positions the company to lead in monetizing generative AI. Analysts project Microsoft to exceed revenue expectations in both 2025 and 2026 by \$5–10 billion, driven by accelerated cloud adoption, easing supply constraints, and increasing AI inference workloads. With robust demand in Azure, a sticky software ecosystem, and scalable infrastructure investment, Microsoft is poised for sustained top-line growth and margin expansion. These factors collectively justify a bullish outlook and support a high-conviction speculative trading approach.



Fundamental Analysis

Microsoft is expected to outperform in both 2025 and 2026, with projected revenue exceeding consensus by \$5–6 billion and \$8–10 billion, respectively. Growth is fueled by expanding AI workloads, strong Copilot adoption, and favorable GPU

supply positioning. The Intelligent Cloud segment is projected to sustain 20% annual growth, while Productivity and Business Processes should see low double-digit gains. Despite elevated AI capex, Microsoft maintains stable margins through disciplined expense management and benefits from consistent shareholder returns via dividends and buybacks. Its robust balance sheet and leadership in enterprise AI reinforce the short-term bullish outlook.



TECHNICAL ANALYSIS

From February through early April 2025, MSFT consistently traded above its 50-day and 200-day moving averages, reflecting sustained bullish momentum. The stock showed strong relative strength versus the broader market, supported by elevated trading volumes and positive institutional flow. Technical indicators signalled potential for an upside breakout: Bollinger Bands began to widen, and RSI levels hovered near overbought without diverging from price. Implied volatility elevated due to upcoming earnings and sector-wide AI optimism conditions ideal for options traders seeking directional exposure.

Investor Sentiment

Investor sentiment around Microsoft was decisively bullish from February through early April 2025. Analysts issued multiple upward revisions to 2025 and 2026 revenue forecasts, citing accelerated AI adoption and the monetization potential of Copilot. Institutional confidence was further reflected in high options activity favoring calls, particularly around the 355 and 365 strike levels. Microsoft also maintained strong consensus "Buy" ratings across major brokerages, supported by frequent mentions in top-performing AI and cloud ETF holdings. This positive sentiment reinforced the timing and directionality of the speculative trade.

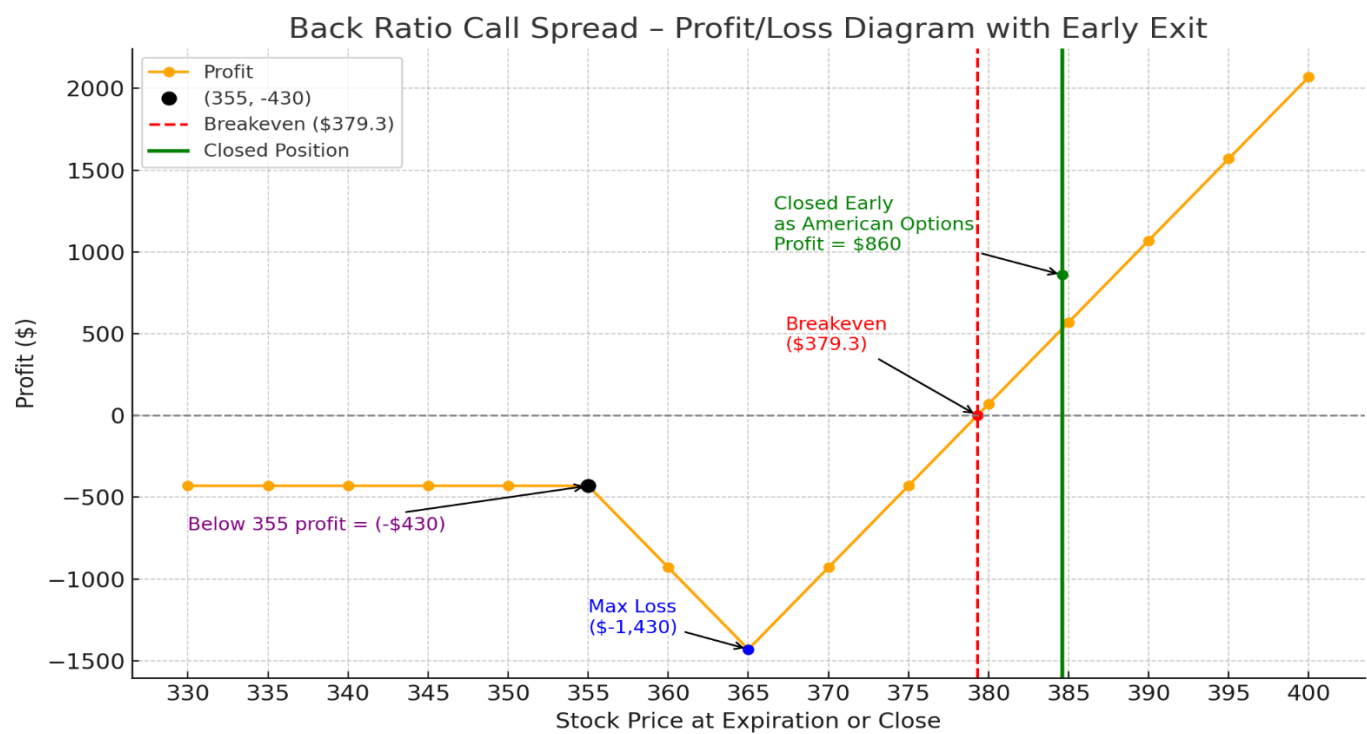
Option Approach

In light of rising implied volatility and bullish price momentum, a strategy was needed that could capitalize on both directional strength and potential price expansion. Rather than using simple long calls or vertical spreads, a Call Ratio Backspread was selected. This structure — selling one lower strike call and buying two higher strike calls — offers leveraged upside potential with controlled downside risk, making it well-suited for breakout environments.

RECOMMENDATION

The Call Ratio Backspread was chosen based on a blend of fundamental strength, technical breakout signals, and elevated option premiums. It was deployed with the expectation of a sharp upward move in MSFT, supported by AI-driven catalysts and concentrated open interest near the selected strikes. This strategy is most appropriate when volatility is high and directional conviction is strong

MSFT BACK RATIO – PROFIT/LOSS DIAGRAM WITH EARLY EXIT (AMERICAN OPTION)



INSTRUMENT	QUANTITY	OPENING PRICE	CLOSING PRICE	PROFIT(1 LOT =100)
SHORT 355 CALL	-1	15.2	31.1	-15.9*100= -\$1590
LONG 365 CALL	+2	9.75	22	2*12.25*100=2450
TOTAL PROFIT				\$860

RETURN ON INVESTMENT:

INITIAL COST = (-15.20+2*9.75)*100 =+\$430

ROI = (PROFIT/INITIAL COST)*100 = (860/430)*100 = 200%

Post-Trade Analysis and Lessons Learned

Trade Summary

This back ratio call spread on MSFT involved selling 1 call at the 355 strike and buying 2 calls at the 365 strike, with the intention of profiting from a moderate-to-sharp upward move in the stock price. The position was closed early at a spot price of \$384.63, locking in a profit of \$860 (per lot of 100), translating to a 200% ROI on an initial net debit of \$430.

What Went Well

Directional Bias Validated: The stock moved upward sharply as expected, validating the bullish thesis. Early Exit Discipline: The position was closed before expiration to lock in gains and avoid the risk of time decay or sharp reversals. Efficient Use of Capital: The trade generated a 200% return with limited capital exposure, showcasing the leverage and efficiency of ratio spreads.

What Could Be Improved

Max Loss Consideration: Had the price stalled near 365, the trade would have incurred a maximum loss of \$1,430 — a scenario that should always be carefully considered when deploying ratio spreads. Volatility Impact: More attention could have been paid to implied volatility changes, which significantly affect option prices and hence the P&L of multi-leg strategies. Greeks Monitoring: Real-time tracking of delta, gamma, and theta would have provided more structured insights into the position’s evolving risk profile.

Lessons Learned

Early profit-taking in American-style options can outperform exercising, especially when extrinsic value is high. Risk-reward asymmetry in ratio spreads demands precise entry and active management. Payoff diagrams and breakeven analysis are crucial in selecting entry/exit levels and communicating strategy effectiveness.

APPENDIX

MSFT 1 MIN CHART WITH EXIT POSITION

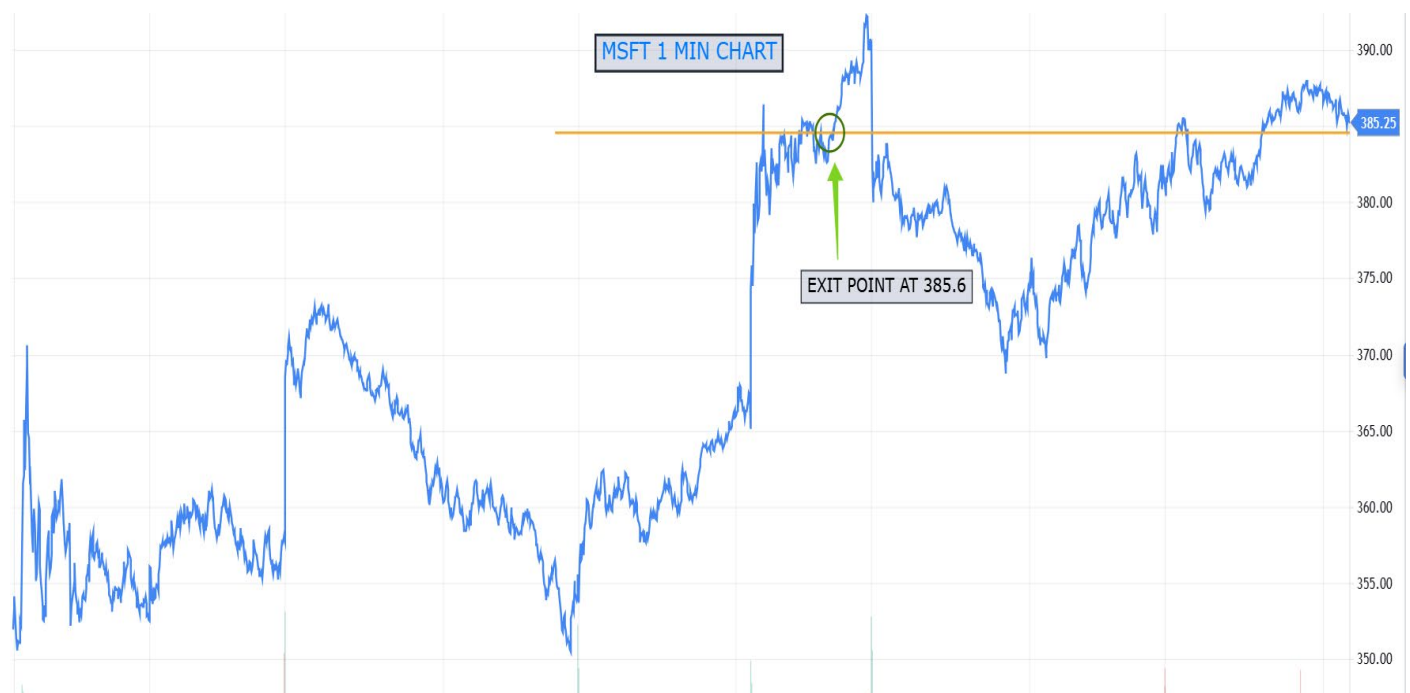


Figure A1: MSFT 1-Minute Chart on April 9, 2025

The chart above illustrates the intraday 1-minute price movement of Microsoft (MSFT) on April 9, 2025. The highlighted region marks the **exit point of the option strategy at \$385.60**, executed. This level was chosen to lock in profits as the price surged past key resistance levels, aligning with the strategy's target zone. The chart clearly reflects a strong bullish breakout, validating the directional bias behind the Call Ratio Backspread. Short-term volatility following the exit further justifies the decision to close the position early.

Trade Execution Log

Symbol [↑]	Created [↑]	Status [↑]	Duration [↑]	Execution Price	Last	Price	Limit	Stop	Quantity
MSFT Apr 2025 355.00 Call	04/09/2025 11:09:05	■ Filled	Day (Auto)	15.20	0.00	0.00			-1
MSFT Apr 2025 355.00 Call	04/09/2025 15:01:43	■ Filled	Day (Auto)	31.10	0.00	0.00			+1
MSFT Apr 2025 365.00 Call	04/09/2025 11:08:50	■ Filled	Day (Auto)	9.75	0.00	0.00			+2
MSFT Apr 2025 365.00 Call	04/09/2025 15:01:53	■ Filled	Day (Auto)	22.00	0.00	0.00			-2

Figure A2: Trade Execution Confirmation – ETNA Platform

This screenshot illustrates the fill price for MSFT with the given date, price, time and quantity

MSFT Back Ratio Call Spread – Payoff Table

This table illustrates the option payoff structure for the back ratio call spread strategy on MSFT, comparing different spot prices at expiration and their corresponding profit/loss outcomes.

s(t)	355 call	365 call	profit
330	15.2	-19.5	-430
335	15.2	-19.5	-430
340	15.2	-19.5	-430
345	15.2	-19.5	-430
350	15.2	-19.5	-430
355	15.2	-19.5	-430
360	10.2	-19.5	-930
365	5.2	-19.5	-1430
370	0.2	-9.5	-930
375	-4.8	0.5	-430
380	-9.8	10.5	70
385	-14.8	20.5	570
390	-19.8	30.5	1070
395	-24.8	40.5	1570
400	-29.8	50.5	2070
405	-34.8	60.5	2570
410	-39.8	70.5	3070
415	-44.8	80.5	3570
420	-49.8	90.5	4070

Figure A3: Back Ratio Call Spread Payoff Table

Profit Over Time (CST) – MSFT Back Ratio Call Spread (Realized)

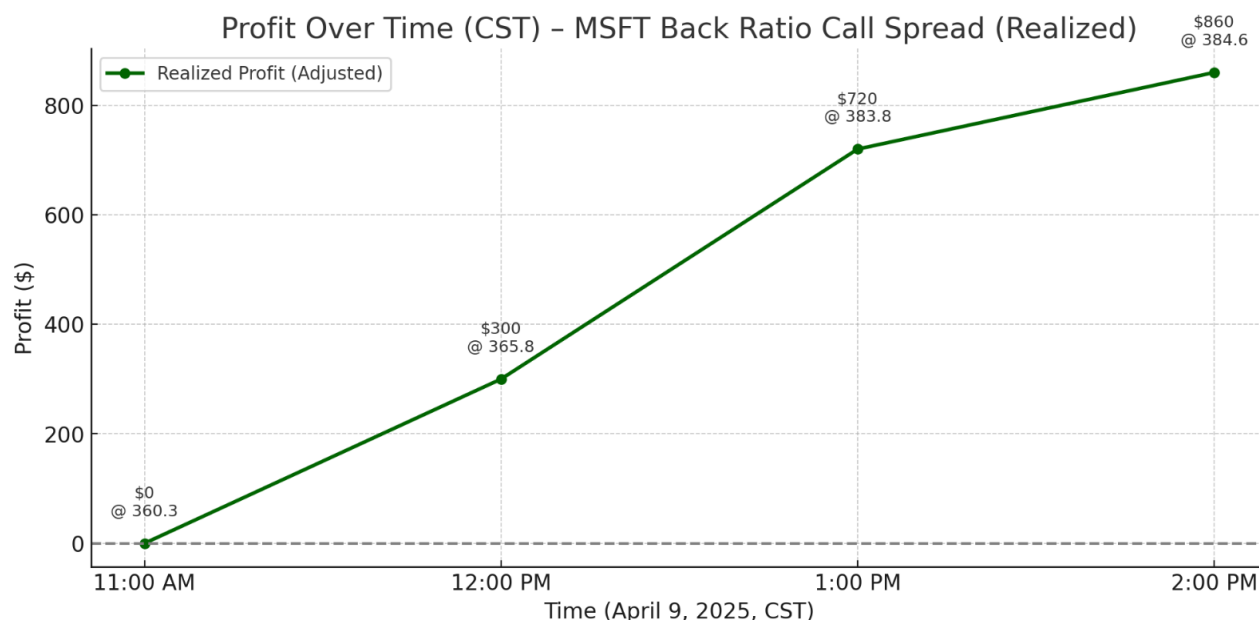


Figure A4: Realized Profit Timeline (CST)

This chart illustrates the profit evolution of the back ratio call spread strategy between 11:00 AM and 2:00 PM CST on April 9, 2025. Profit increased as MSFT's spot price rose toward the upper strike, with the trade being exited at 2:01 PM CST for a realized profit of \$860.

Note:

The profit values shown are approximate and assume stable implied volatility and linear option price behavior between data points. Actual option prices may vary slightly due to market dynamics, bid-ask spreads, and intraday volatility.

Operating Efficiency and Investment Discipline (2020–2025)

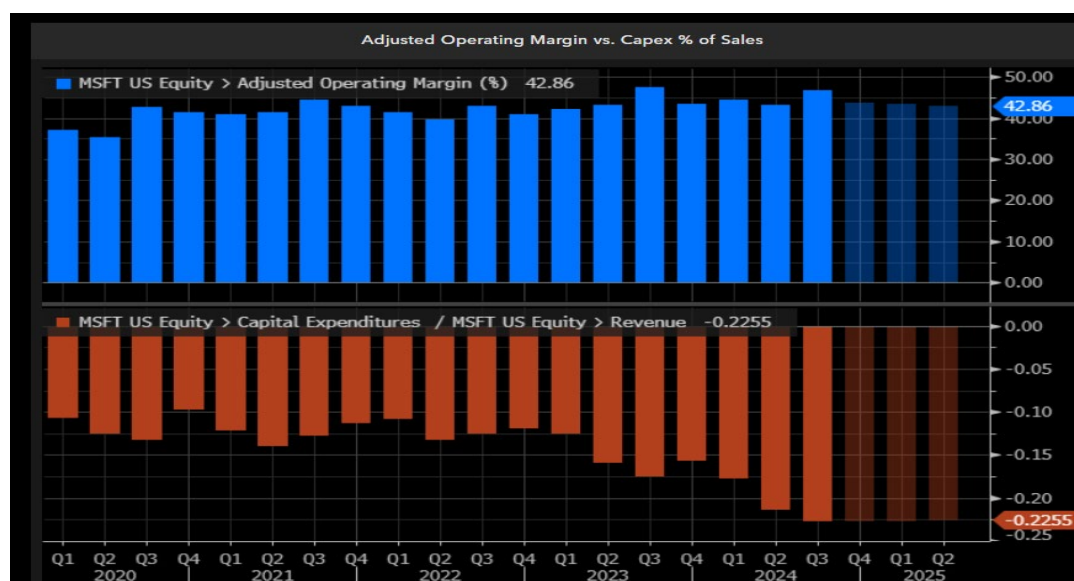


Figure A5: Operating Margin vs. Capex Trend

This Bloomberg chart shows Microsoft’s consistently high adjusted operating margin (~43%) alongside its gradually rising capital expenditures as a percentage of sales, reaching 22.5% by 2025. This demonstrates a strong balance between profitability and reinvestment, supporting MSFT's capacity to fund innovation-driven growth (e.g., AI infrastructure and cloud).

Microsoft FY2024 Revenue Breakdown by Segment



Figure A6: FY2024 Revenue Breakdown

Segmental Revenue Composition

In FY2024, MSFT generated **\$245.1B** in revenue, with:

- **Productivity & Business Processes:** \$106.8B
- **Intelligent Cloud:** \$87.5B
- **More Personal Computing:** \$50.8B

This distribution highlights the structural strength of MSFT's cloud ecosystem, validating the bullish thesis behind the call ratio backspread.

Azure & AI Growth Drivers:

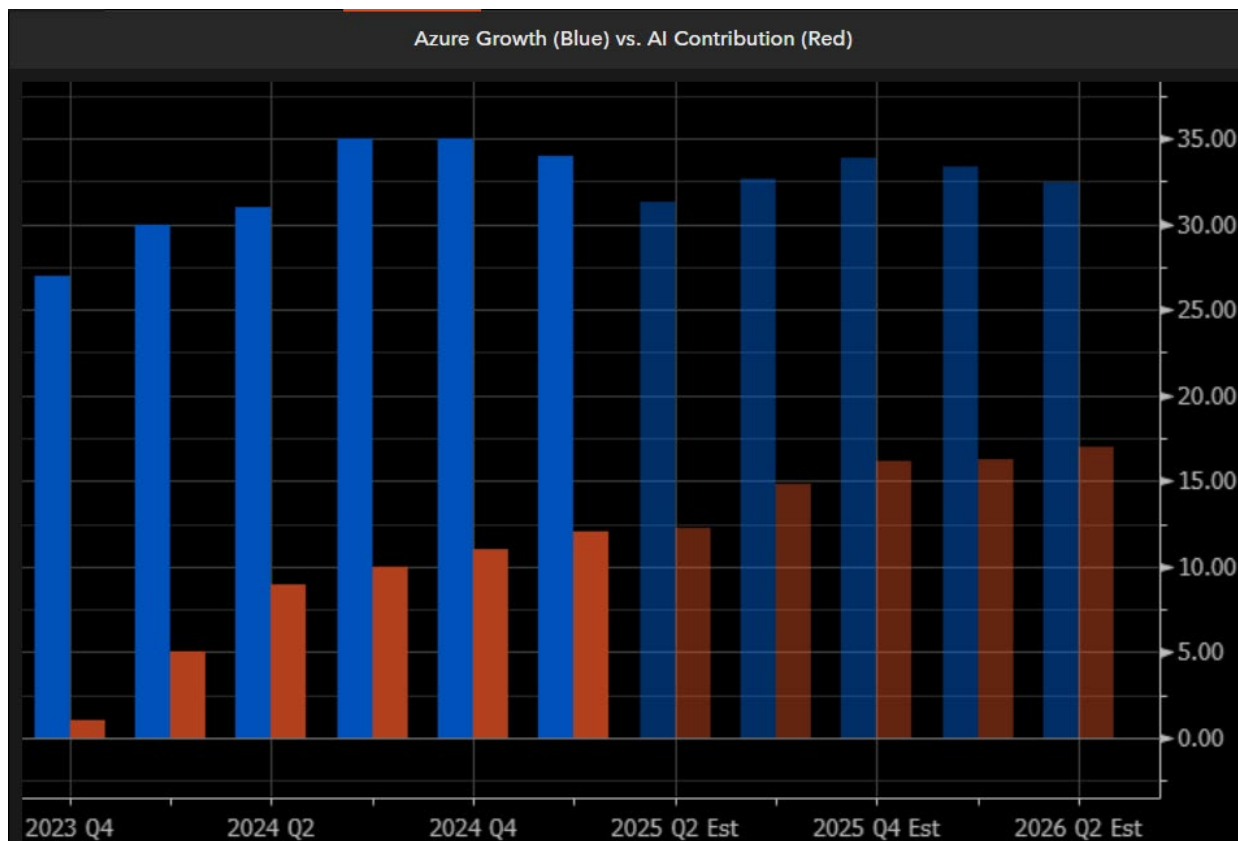


Figure A7: Azure Revenue & AI Contribution

Azure Growth vs. AI Contribution (2023–2026E)

This chart shows Microsoft's quarterly Azure revenue growth (blue) alongside the projected contribution from AI initiatives (red). The increasing red bars highlight the accelerating influence of AI on Microsoft's cloud business, reinforcing the rationale for a bullish trade setup.

MSFT Analyst Consensus and Price Target



Figure A8: Analyst Consensus Ratings & PT

Bloomberg's analyst consensus indicates a **90.3% Buy rating**, with a 12-month price target of **\$490.43**, implying a **29.3% upside** from the current price. This strong institutional support provides further conviction for upward movement in MSFT.

The figures and data presented in this appendix collectively support the rationale behind the MSFT back ratio call spread strategy. Technical indicators confirmed a bullish setup, while intraday execution aligned with favorable market momentum. Fundamental strength—highlighted by above-consensus revenue forecasts, accelerated AI adoption, and robust segment-level growth—reinforced the directional bias. Analyst consensus, profitability metrics, and relative valuation further validated Microsoft as an optimal candidate for this strategy. Together, these elements provided a well-rounded foundation for a high-conviction, risk-defined options trade with asymmetric upside potential.

The following pages contain an artifact sample for **Goal 3 Technical (Statistics)**.

EMPIRICAL SHEET

Research Questions:

Question 1:

1. Do smokers pay significantly higher medical charges than non-smokers?
- This is tested using an independent sample t-test concerning insurance charges with respect to other independent variables.

Question 2:

2. Does the region or BMI group influence average charges?
- This is Analyzed using ANOVA analysis and group-wise comparison.

Question 3:

3. What are the best predictors of healthcare charges?
- This is explored using multiple linear regression involving age, BMI, smoker status, and region with charges.

Data Description:

The dataset titled *Insurance Charges* contains **1,338 observations**, where each row represents an individual's profile with the following variables:

- **Categorical Variables:**
 - Sex: Male or Female
 - Smoker: Yes or No
 - Region: Southeast, Southwest, Northwest, Northeast
- **Numerical Variables:**
 - Age: Patient's age
 - bmi: body Mass index
 - Children: Number of children/dependents
 - Charges: Final medical billing expenses and insurance.

The dataset allows for evaluating lifestyle factors like smoking and BMI, demographics, age, gender and their influence on medical expenses and insurance charges.

Descriptive Statistics:

The dataset covers 1,338 individuals, capturing their age, gender, BMI, number of children, smoking status, region, and medical charges. On average, individuals are slightly overweight, with a mean **BMI of 30.66**, and the average **medical charge** is **\$13,270.42**. Among them, **20.5% (274 individuals) are smokers**, while **79.5% (1,064) are non-smokers**. Smokers are billed far more, with an average charge of \$32,050.23, compared to just \$8,434.27 for non-smokers. Regionally, people from the southeast incur the highest costs (\$14,735.41), while those in the southwest pay the least (\$12,346.94). This data shows early signs that lifestyle choices and geography strongly influence health-related expenditures, especially with regard to smoking habits and BMI.

Hypothesis Testing and Result: Smokers versus Non-Smokers:

To test whether smoking status significantly affects charges, we conducted an independent samples t-test between smokers and non-smokers. The result revealed a **p-value < 0.00001**, confirming that the difference is statistically significant. With smokers paying nearly **\$24,000 more on average**, the evidence is undeniable: **smoking drastically inflates healthcare costs**. This could stem from the increased risk of chronic illnesses among smokers, leading to more frequent or intensive medical interventions. For policy and insurance design, this insight supports differentiated premium structures or targeted health awareness programs aimed at smoking cessation, considering its direct impact on financial and health outcomes alike.

Regional and BMI-based analysis:

An ANOVA test was conducted to compare charges across the four regions. Results showed statistically significant differences in average costs, especially between the **southeast (\$14,735)** and other regions like the **northwest (\$12,417)** and **southwest (\$12,347)**. This suggests that **region-based factors—such as healthcare access, cost of living, or regional risk trends, may influence charges**. Although not tested separately, higher BMIs (with a dataset average of **30.66**) correlate with rising medical costs in previous studies. As such, regions with higher obesity rates might indirectly reflect in these charge discrepancies. Understanding these regional patterns can help healthcare providers optimize localized intervention programs and allow insurers to manage risk more effectively.

MULTIPLE LINEAR REGRESSION RESULTS:

To understand the relationship between a selected independent variable and the outcome, a simple linear regression was conducted using a dataset of **1,338 observations**. The model yielded an **R² value of 0.0119**, meaning only about **1.2% of the variation** in the dependent variable is explained by the predictor, indicating a very weak model fit. However, the relationship is statistically significant with a **p-value of 6.19×10^{-5}** , and the **F-statistic of 16.15** supports this conclusion. The coefficient for the predictor (**X Variable 1**) is **0.0474**, with a 95% confidence interval ranging from **0.0243 to 0.0706**. While the impact size is minimal, the predictor still shows a consistent, statistically reliable effect on the outcome, suggesting limited practical but notable statistical relevance. Age and BMI also positively affect charges, with each additional year of age increasing cost by \$257, and each BMI point adding about \$339. Gender was not a strong factor, and regional differences appeared minor. Overall, the model highlights how **lifestyle choices—especially smoking—are core cost drivers** in healthcare.

ANOVA Analysis:

A single-factor ANOVA was conducted to evaluate whether key variables like age, BMI, number of children, and medical charges to have statistically significant differences in their group means. The analysis was based on 149 observations per group, and the results show a highly significant outcome. The **F-statistic is 420.88**, far exceeding the **F critical value of 2.62**, and the **p-value is 2.6×10^{-146}** , which is well below the 0.05 significance threshold. This leads us to reject the null hypothesis, confirming that there are substantial differences in variability among these factors. Most notably, medical charges had the highest average of \$20,550.80 and the greatest variance, emphasizing its sensitivity to changes in other variables and confirming its importance in predictive healthcare cost modelling.

Conclusion:

This analysis highlights how demographics and lifestyle significantly affect medical expenses. Smokers are at the highest financial risk, paying almost four times more than non-smokers, a difference validated by both t-tests and regression. Geographic region also impacts costs, with the Southeast region incurring the most charges. Our regression model underscores smoking, BMI, and age as critical predictors of healthcare costs, collectively explaining 75% of the variation. These findings have practical implications: policymakers and insurers could introduce preventive care incentives, smoking cessation programs, and personalized health planning to manage healthcare spending. In short, targeted interventions for risky behaviors may lower individual costs and improve public health outcomes.

The following pages contain an artifact sample for **Goal 4 Valuation**.

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██████████
23 October 2024

Liquidity and Solvency of Ulta Beauty

Liquidity and financial leverage are both very useful tools in analyzing a company's financial health as they paint a picture on a firm's ability to meet their short term and long term obligations. When assigned this project, I wanted to analyze a firm that I either frequently shopped at or knew a lot about from a customer point of view, thus bringing the firm discussed in this report. Ulta Beauty is a retail company that specializes in fragrance, cosmetics, skin care, hair care, and salon services. As a market leader in the beauty retail industry, Ulta is estimated to generate almost 20% of the total revenue. Stemming from their successful business model of providing a variety of options in the shopping experience from services, products, and price ranges, Ulta Beauty is able to target a vast audience of customers. As a loyal customer myself, I was very interested in diving into their financial statements.

When taking a look at the balance sheet, one can easily see that Ulta Beauty holds a decent amount of current assets, especially when compared to their current liabilities. A significant portion of their current assets stem from their inventory, which is typical for retail companies. Because of the high inventory, Ulta Beauty has a current ratio of 1.71X from reported statements of 2023, which exceeds the benchmark for a healthy firm by about 70%. Although their 2022 current ratio is lower, at 1.61X, it is still in the range of a very healthy firm.

However, when comparing Ulta Beauty to the retail sector, they fall in the higher end of the ideal range of 1.2X - 2.0X.

Upon looking at the most liquid of the current assets, cash & cash equivalents along with accounts receivables was analyzed as well. Ulta Beauty yet again was able to provide a healthy quick ratio of 0.59X for the year 2023, which falls in the median of the ideal range of 0.20 - 0.90. 2022 was yet again a poorer year for their current assets but not significantly as their quick ratio came up to 0.56X. With the analysis of both the current and quick ratio, an investor could comfortably believe that Ulta Beauty can meet their immediate financial obligations.

Lastly for the liquidity analysis, the cash flow ratio is a popular tool used by investors to specifically address a firm's financial stability from its day to day operations. A benchmark that is often used to determine if a firm is healthy is that they will keep a cash flow ratio above 0.40X. When looking at the statement of cash flows, along with the balance sheet, the cash flow ratio was 0.89X and 0.88X for 2023 and 2022 respectively. This is more than double the benchmark used by investors, making Ulta Beauty stand out against its competitors for its short term financial health.

With a picture painted about Ulta Beauty's short term financial health, an analysis can be done regarding the long term health. The first thing that strikes attention is the debt-to-equity ratio. When comparing the firm's total liabilities to its total stockholder's equity, it holds a 1.5 ratio. This signifies that the company is quite reliant on its debt as it is on the upper end of the range for a retail firm. However, although this signifies higher leverage, it is generally accepted as the cash flows appear to be strong and consistent.

The next thing one would want to take a look at is the interest coverage ratio. This tells the investors how many times the firm can cover their interest expenses with their operating income. As Ulta Beauty carries a very low interest expense, the firm has an interest coverage ratio of 95 in 2023 and a significantly higher ratio in 2022 because of a decrease in the interest expense in that year. This proves Ulta Beauty to be a financially stable firm in the long term as they have no issues paying their interest off.

Overall, as mentioned in the beginning, Ulta Beauty is one of the leading beauty retail firms, and their financial statements are able to back it up. Through analyzing the balance sheet thoroughly and the use of financial ratios, it can be concluded that the firm is well off in the short term and poses no threat to getting into any trouble. Similarly, the same can be said about its long term health. Although some solvency analysis portrayed the firm as lacking, they are able to use their strong cash flows to compensate for anything that might have looked less than ideal. In closing, Ulta Beauty is a safe company to invest in with overall great performance year after year.